Mastering Data: SQL Functions

Built-in tools that transform raw database records into meaningful insight.

The Power of SQL Functions

What They Are

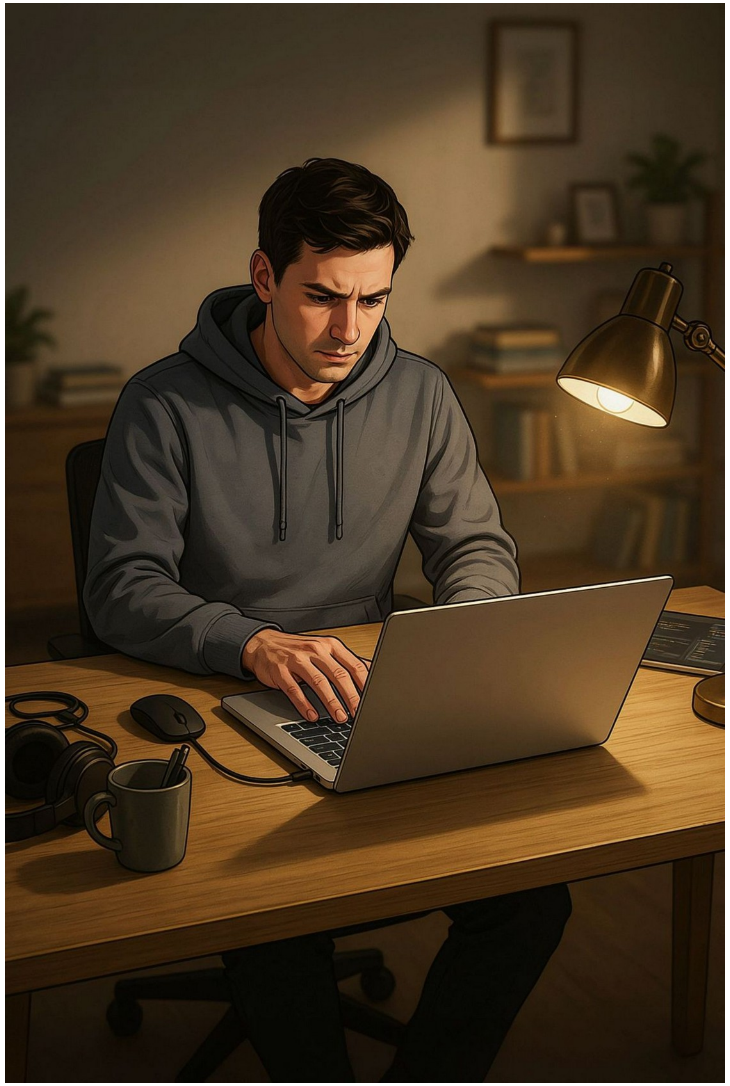
Built-in operations that process data and return a single result

What They Do

Calculate, format, and transform data directly inside your queries

Why They Matter

Eliminate manual processing — let the database do the heavy lifting



Aggregate Functions

SUMMARIZING DATA

What Are Aggregate Functions?

Operate on a **set of values** to produce one summary result per group.

- Essential for **analytical reporting**
- Identify **trends and patterns** across large datasets
- Power dashboards, KPIs, and **business intelligence**

Persistent

The Core Aggregate Toolkit

COUNT()

Total number of rows or non-null values in a column

SUM()

Calculates the total of all values in a numeric column

AVG()

Computes the mean value across a numeric set

Use these in `SELECT` statements with or without `GROUP BY` to slice your summaries by category.

Precision in Aggregation

MIN()

Returns the **smallest value** in a column — useful for floor prices, earliest dates, lowest scores

MAX()

Returns the **largest value** — useful for peak sales, latest timestamps, highest ratings

⚠ Important Note on NULLs

All aggregate functions **automatically exclude NULL values** during calculation. Always account for missing data in your results.

```
SELECT COUNT(salary)
-- Skips NULL rows
```

String Functions: Mastering Text

UPPER()

Converts all characters to **UPPERCASE** — normalize input for consistent searches

LOWER()

Converts all characters to **lowercase** — standardize display and comparisons

Formatting consistency starts at the query level — never rely on the application layer alone.

Persistent

Advanced String Manipulation

CONCAT()

Joins two or more strings into one meaningful label.

```
SELECT CONCAT(first_name,
  ' ', last_name)
AS full_name
FROM employees;
```

SUBSTRING()

Extracts a portion of a string by **position** and **length**.

```
SELECT SUBSTRING(phone,
  1, 3)
AS area_code
FROM contacts;
```

Combine CONCAT and SUBSTRING to build formatted outputs like masked card numbers or structured name fields.

Date Functions: Managing Time

NOW()

Returns the **current date and time** as a full timestamp — ideal for logging created_at or updated_at fields

CURDATE()

Returns **today's date only**, without the time component — perfect for date-range filters and comparisons

Extracting Date Components

Breaking Down a Date

Use `YEAR()` and `MONTH()` to isolate specific parts of any date field for period-based analysis.

```
SELECT
  YEAR(order_date) AS yr,
  MONTH(order_date) AS mo,
  COUNT(*) AS total_orders
FROM orders
GROUP BY yr, mo
ORDER BY yr, mo;
```

YEAR()

Annual trend reporting, year-over-year comparison

MONTH()

Seasonal analysis, monthly KPIs, period grouping

Turning Data into Decisions

Combine these functions to **extract, summarize, and format** complex datasets in a single query.

1

Aggregate

Summarize with COUNT, SUM, AVG, MIN, MAX

2

Format

Clean and shape text with UPPER, LOWER, CONCAT, SUBSTRING

3

Contextualize

Add time intelligence with NOW, CURDATE, YEAR, MONTH

Master these functions and you'll transform raw database records into clear, actionable business intelligence.

Persistent

Persistent

Thank you